

MASHHAD INTL, Iran, Islamic Republic of

WMO: 407450

Lat: 36.2366N Lon: 59.6317E Elev: 3278 StdP: 13.04 Time Zone: 3.50 (IRN) Period: 94-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	21.5	4.8	7.9	37.8	10.0	10.3	42.2	21.1	42.0	17.9	43.2	3.2	150	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	100.1	63.6	97.2	63.1	95.2	62.4	70.2	91.7	68.1	90.0	66.1	87.9	12.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	95.9	83.1	59.4	85.6	79.3	57.5	79.6	77.3	36.3	92.4	34.3	90.2	32.7	88.0	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature										
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years				
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max			
1%	2.5%	5%															
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	17.4	15.2	DB	12.7	104.2	7.2	2.2	7.5	105.8	3.3	107.1	-0.7	108.3	-6.0	109.9		
			WB	11.9	70.7	7.2	3.5	6.7	73.2	2.5	75.2	-1.5	77.2	-6.7	79.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	38.1	41.4	50.2	60.1	70.6	79.3	82.9	79.9	71.8	61.0	49.0	41.1
	DBStd	17.20	8.60	8.85	8.96	8.67	6.45	4.73	3.93	4.19	5.64	7.45	7.94	7.09
	HDD50	1163	374	257	109	20	1	0	0	0	0	9	109	284
	HDD65	3569	833	661	462	189	23	1	0	0	12	165	482	741
	CDD50	5019	5	16	115	323	640	879	1020	928	655	351	79	8
	CDD65	1951	0	0	3	42	197	430	555	463	217	42	2	0
	CDH74	24416	0	2	63	498	2084	5364	7485	5914	2413	559	34	0
	CDH80	12495	0	0	9	143	842	2843	4261	3199	1032	161	5	0
(14) Wind	WSAvg	6.2	4.6	5.9	6.4	6.7	6.8	7.6	8.1	7.6	6.3	5.3	4.4	4.2
(15) Precipitation	PrecAvg	8.6	0.9	1.3	2.0	1.6	1.1	0.1	0.1	0.0	0.1	0.4	0.7	0.9
	PrecMax	12.2	3.1	5.1	4.6	5.1	4.6	1.9	0.7	0.2	1.0	2.3	2.4	3.1
	PrecMin	4.6	0.0	0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.4	0.7	1.1	1.1	1.1	1.0	0.3	0.2	0.1	0.2	0.6	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.8	69.4	80.2	89.6	96.5	101.9	103.7	102.0	96.9	89.8	78.7	66.5
		MCWB	46.0	49.4	54.9	59.7	61.4	63.8	64.5	63.9	60.7	59.4	55.1	47.2
	2%	DB	59.4	64.1	74.9	84.1	91.7	98.5	100.5	98.5	93.1	84.3	71.5	61.0
		MCWB	44.2	47.8	52.4	59.3	60.7	62.6	63.7	63.5	60.3	57.6	51.3	45.0
	5%	DB	55.4	59.4	70.0	80.1	88.1	95.3	98.4	96.2	89.6	80.3	66.6	57.1
		MCWB	42.7	45.3	51.1	57.6	60.3	61.5	63.7	63.3	59.6	56.1	49.6	43.9
	10%	DB	51.4	55.5	66.0	75.4	84.5	93.0	95.3	93.3	86.1	76.0	62.7	53.3
		MCWB	40.9	44.0	49.8	55.8	59.2	61.4	63.3	62.0	58.8	54.5	48.1	42.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	52.3	58.2	63.7	67.5	69.5	74.3	73.4	69.0	62.0	56.7	49.4
		MCDB	58.7	62.1	72.3	80.8	84.9	93.3	96.9	91.9	89.8	81.7	73.5	61.8
	2%	WB	45.9	49.5	55.0	61.2	64.3	67.3	70.8	70.0	65.8	59.2	53.8	47.1
		MCDB	56.8	59.8	67.9	77.7	82.5	89.4	92.5	91.4	85.3	79.0	66.5	57.9
	5%	WB	43.8	47.2	52.8	59.5	62.6	65.5	68.3	67.5	62.8	57.4	51.7	45.0
		MCDB	53.2	56.7	65.7	74.5	80.5	87.3	90.6	89.7	82.1	76.1	62.5	54.4
	10%	WB	41.5	45.1	51.0	57.6	61.2	63.8	66.2	64.8	60.3	55.7	49.5	43.2
		MCDB	49.8	53.8	63.4	71.8	79.0	85.9	88.7	87.3	80.4	72.6	60.1	51.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	18.8	19.9	21.2	23.6	25.1	24.4	26.1	26.6	24.9	20.6	18.4
		MCDBR	25.3	25.5	27.5	28.4	28.4	28.5	27.2	28.8	30.6	30.5	27.4	25.5
		MCWBR	13.1	12.4	11.2	10.9	9.9	9.5	9.3	11.1	12.0	13.0	12.3	13.0
	5% WB	MCDBR	22.3	22.1	24.1	24.2	24.9	25.0	23.7	26.3	27.0	26.6	23.2	21.9
		MCWBR	12.4	11.8	11.1	10.4	9.7	10.8	11.7	14.0	13.4	12.1	11.1	11.9
(40) Clear-Sky Solar Irradiance	taub		0.335	0.378	0.454	0.498	0.516	0.494	0.486	0.440	0.404	0.423	0.381	0.333
	taud		2.278	2.139	1.938	1.876	1.858	1.907	1.934	2.045	2.118	2.045	2.148	2.292
	Ebn at Noon		268	267	255	252	249	253	255	264	268	247	246	260
	Edh at Noon		32	42	56	62	65	62	60	52	46	45	36	30
(44) All-Sky Solar Radiation	RadAvg		790	1039	1386	1808	2195	2546	2519	2357	1961	1407	944	739
(45)	RadStd		70	127	108	169	113	85	56	70	55	73	80	70

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	+0.77	N/A	-3.24	+0.61	-3.26	-2.86	N/A	N/A	N/A	+160
Regional (7 neighbors)	N/A	N/A	N/A	+0.90	-1.33	-1.30	N/A	N/A	N/A	+158